Towards Multi-Granularity Memory Association and Selection for Long-Term Conversational Agents

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Abstract

Large Language Models (LLMs) have recently been widely adopted in conversational agents. However, the increasingly long interactions between users and agents accumulate extensive dialogue records, making it difficult for LLMs with limited context windows to maintain a coherent long-term dialogue memory and deliver personalized responses. While retrieval-augmented memory systems have emerged to address this issue, existing methods often depend on single-granularity memory segmentation and retrieval. This approach falls short in capturing deep memory connections, leading to partial retrieval of useful information or substantial noise, resulting in suboptimal performance. To tackle these limits, we propose MemGAS, a framework that enhances memory consolidation by constructing multi-granularity association, adaptive selection, and retrieval. MemGAS is based on multi-granularity memory units and employs Gaussian Mixture Models to cluster and associate new memories with historical ones. An entropy-based router adaptively selects optimal granularity by evaluating query relevance distributions and balancing information completeness and noise. Retrieved memories are further refined via LLM-based filtering. Experiments on four long-term memory benchmarks demonstrate that MemGAS outperforms state-of-the-art methods on both question answer and retrieval tasks, achieving superior performance across different query types and top-K settings.[1]

1 Introduction

Large Language Models (LLMs) have showcased remarkable conversational abilities, enabling them to serve as personalized assistants [22, 47, 23] and handle a wide range of applications, such as customer service [34] and software development [36]. However, as user-agent interactions increase, the volume of conversational history grows significantly. Despite their impressive conversational abilities, LLMs struggle with maintaining long-term conversational memory due to their limited context length [27], which makes it challenging to retain a comprehensive record of user interactions and preferences over time. This limitation undermines their ability to generate coherent and personalized responses [50, 33]. The development of retrieval-based external (non-parametric) memory systems [32, 35] has emerged as a promising solution. By storing interaction histories and retrieving relevant information when needed, memory systems enable LLMs to recall user-specific details from past dialogues and deliver more tailored responses.

Recent studies on external memory systems of LLM agents primarily depend on the Retrieval-Augmented Generation (RAG) pipeline and have explored diverse aspects of memory segmentation

[1]https://github.com/quqxui/MemGAS

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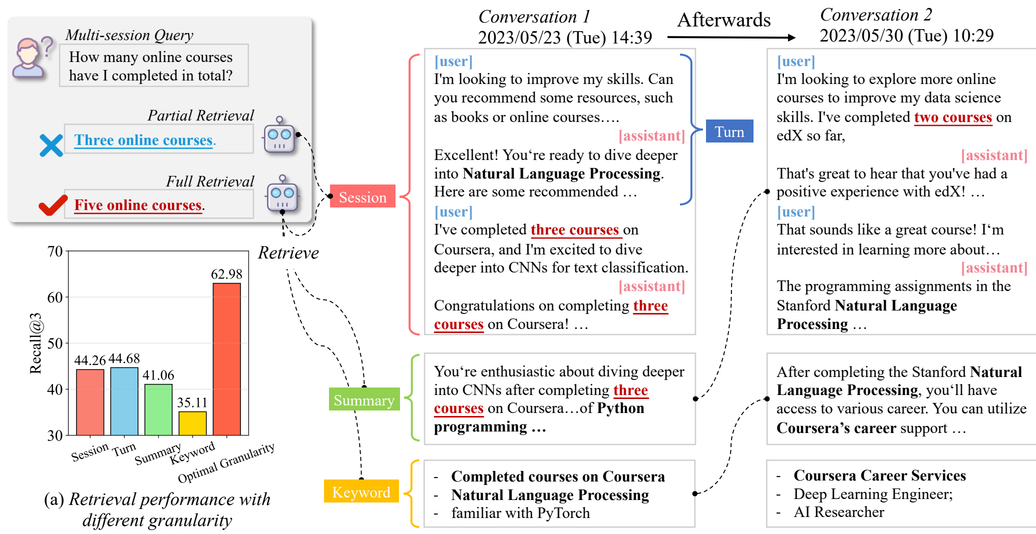

Figure 1: An example of the multi-session query. The agent needs to synthesize information from both *Conversation 1* and *Conversation 2* to answer the question. The **red underlines** highlight information directly relevant to the answer, while the **bold text** emphasizes the same details between conversations. By generating multi-granularities (e.g., summary and keyword), the agent can better establish connections between conversations, enhancing full memory retrieval. Additionally, chart (a) illustrates the performance across different granularities, where 'optimal granularity' means the result using best-suited granularity for each query.

and construction for effective retrieval [51, 56]. For memory segmentation, existing approaches mainly adopt single-granularity to segment conversations. They utilize session-level chunks as retrieval units [28, 26, 45], while others employ finer-grained turn-level segmentation to capture details [50, 54]. Recent advances introduce topic-aware segmentation techniques [33, 44] that group dialogue based on semantic coherence, enhancing topic-consistent retrieval. Additionally, several works [49, 45, 59] generate memory summary, condensing key information into compact representations to improve retrieval efficiency. Regarding memory construction, researchers have explored structured organization paradigms to enhance long-term knowledge retention [25, 6, 7]. Methods like RAPTOR [42] and MemTree [39] employ hierarchical tree structures to encode multi-scale relations between memory units. Inspired by cognitive mechanisms [46], HippoRAG [8, 9] implement graph-based memory architectures that simulate neural memory consolidation processes, establishing relations between entities.

However, despite their progress, current approaches exhibit two critical limits: **i) Insufficient Multi-Granularity Memory Connection.** While existing methods endeavor to organize memories into topological structures (e.g., knowledge graphs [8, 9] or trees [42, 39]), they predominantly concentrate on singular granularity levels—either entities or session summaries. This single-scale paradigm fails to model cross-granular interactions between memory units, resulting in retrieving only partial useful information. As demonstrated in Figure 1, answering multi-session queries requires establishing semantic links across granularities (e.g., connecting *Conversation 1* and *Conversation 2* through shared keyword/summary). Failure to establish links results in partial retrieval (e.g., retrieving only *Conversation 1*), which leads to incorrect answers. **ii) Lack of Adaptive Multi-Granular Memory Selection.** Current methods mainly rely on fixed granularity strategies (e.g., session/turn segmentation or LLM-generated summaries [49, 59, 50]), which often lead to incomplete context recall or noise due to improper granularity. Although topic-aware segmentation [33, 44] enhances intra-chunk coherence, they lack adaptive mechanisms to select optimal granularity per query. Our empirical analysis in Figure 1(a) reveals that adaptively choosing optimal granularity per query (e.g., balancing noise reduction in summaries/keywords with information retention in raw sessions) yields substantial performance gains. This highlights the necessity of granularity selection to resolve the inherent noise-information trade-off.

In this paper, we propose MemGAS, a framework for constructing and retrieving long-term memories through multi-granularity association and adaptive selection. Our method addresses these issues by two core strategies: **i) Memory Association:** We leverage LLMs to generate memory summaries and keywords, constructing multi-granular memory units. When new memory is updated, a Gaussian Mixture Model is employed to cluster historical memories into an accept set (relevant) and a reject set (irrelevant). The memories in the accept set are then associated with new memories, ensuring consolidated memory structures and real-time updates. **ii) Granularity Selection:** An entropy-based router adaptively assigns retrieval weights to different granularities by evaluating the certainty of the query's relevance distribution. Finally, critical memories are retrieved using Personalized PageRank and filtered through LLMs to remove redundancies, ensuring a refined and high-quality memory that enhances the assistant's understanding. Experiments on four open-source long-term memory benchmarks demonstrate that MemGAS significantly outperforms state-of-the-art baselines and single-granularity approaches on both question answer (QA) and retrieval tasks. Moreover, it consistently achieves superior results across various query types and retrieval top-k settings.

2 Methodology

This section defines the task and data format, constructs a dynamical memory association framework, details an entropy-based router for optimal granularity, and outlines strategies for retrieving and filtering high-quality contextual information for response generation.

2.1 Preliminary

Our work focuses on building personalized assistants through long-term conversational memory, where the system leverages multi-session user-agent interactions (referred to as memory) to construct an external memory bank $\mathcal{M}$. Without loss of generality, the i-th session $S_i = \{(u_j^{(i)}, a_j^{(i)})\}_{j=1}^{n_i}$ contains n_i dialogue turns of user utterances $u_j^{(i)}$ and assistant responses $a_j^{(i)}$. When the assistant receives a query q, our aim is to retrieve relevant memories $\mathcal{M}_{\text{rel}} \subseteq \mathcal{M}$ through multi-granular associations across session-level S, turn-level T, keyword-level K, and summary-level U, then generates responses via $a = \text{LLM}(q, \mathcal{M}_{\text{rel}})$. The core challenges involve establishing cross-granular memory association and learning adaptive granularity selection $\psi : q \rightarrow \{\alpha_s, \alpha_t, \alpha_k, \alpha_s\}$ to balance information completeness and retrieval noise.

2.2 Multi-Granularity Association Construction

Existing methods mainly encode memories into vector libraries (e.g., session-level chunks) and directly retrieve information via similarity search [13, 19, 28]. However, such approaches overlook deeper associations between memories. To address this, we propose an associative memory construction process that captures multi-granular relationships.

Multi-Granular Memory Metadata. For the i-th session S_i, multi-granularity metadata is generated using LLMs, including a session summary U_i and keywords K_i. Additionally, the session is segmented into multiple turns T_i. Formally,

$$U_i, K_i = f_{\text{LLM}}(S_i), \quad T_i = \text{segment}(S_i) \tag{1}$$

Here, f_{LLM} denotes LLM processing function, and segment represents segmentation operation. The resulting memory chunk M_i combines the session-level S_i, turn-level dialogues T_i, summary U_i, and keywords K_i, and is stored in memory bank as:

$$M_i = \{S_i, T_i, U_i, K_i\} \in \mathcal{M} \tag{2}$$

Dynamical Memory Association. When a new memory $\mathcal{M}_{\text{new}}$ is added, the current memory bank $\mathcal{M}_{\text{cur}}$ is updated as $\mathcal{M}_{\text{cur}} \leftarrow \mathcal{M}_{\text{cur}} \cup \mathcal{M}_{\text{new}}$. To establish associations between $\mathcal{M}_{\text{new}}$ and historical memories, a Gaussian Mixture Model (GMM)-based clustering strategy [38] is used. All memories are encoded into dense vectors $e(\mathcal{M}_{\text{cur}})$ and $e(\mathcal{M}_{\text{new}})$ by Contriever [13], and pairwise similarity scores are computed between $e(\mathcal{M}_{\text{new}})$ and each $e(\mathcal{M}_j) \in e(\mathcal{M}_{\text{cur}})$. The resulting set of similarity vectors $\mathbf{s}_{\text{sim}}$ is clustered by GMM into two probabilistic sets:

- **Accept Set**: Memories with high similarity to $\mathcal{M}_{\text{new}}$, forming direct associations with $\mathcal{M}_{\text{new}}$.

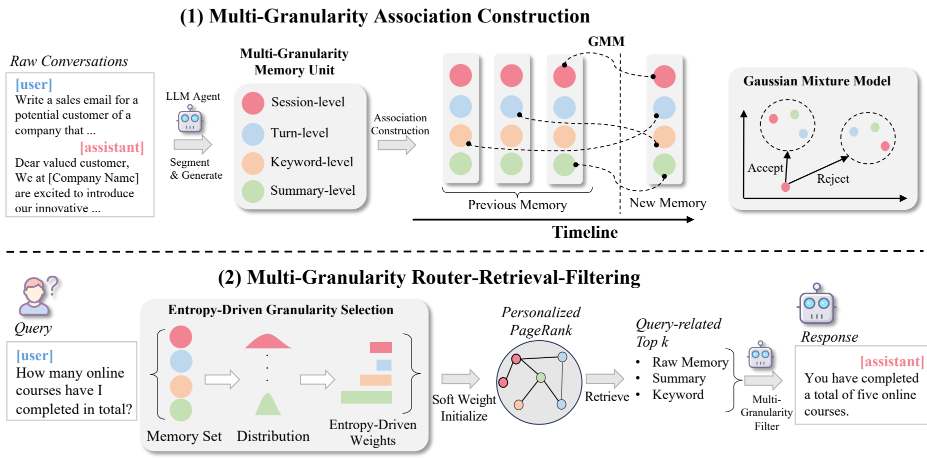

Figure 2: Overall Framework. In the first phase, we leverage LLM agent to generate multi-granularity information and construct Memory Association using GMM. In the second phase, given a query, we perform multi-granularity routing based on entropy and utilize PPR algorithm to search for key nodes. Finally, we filter the information to obtain the answer.

- **Reject Set**: Irrelevant memories, excluded from immediate connections with $\mathcal{M}_{\text{new}}$.

Note that the similarity vectors are computed with granularity-specific information, meaning that each granularity of $\mathcal{M}_{\text{new}}$ is treated as a node and used to establish connections with nodes in $\mathcal{M}_{\text{cur}}$. We maintain an association graph $\mathcal{A}_{\text{cur}}$ to store the connections in $\mathcal{M}$, which is updated as $\mathcal{A}_{\text{cur}} \leftarrow \mathcal{A}_{\text{cur}} \cup \mathcal{A}_{\text{new}}$, where $\mathcal{A}_{\text{new}}$ represents the edges between $\mathcal{M}_{\text{new}}$ and its **accept set**. This process mimics human-like memory consolidation by selectively reinforcing contextually related memories.

2.3 Multi-Granularity Router

Existing methods rely on single predefined granularity for memory retrieval, limiting their ability to adaptively prioritize fine- or coarse-grained information based on query [19, 49, 42]. To address this, we propose an entropy-based router that adaptively selects the optimal granularity for each query.

Entropy-Driven Granularity Selection. For a query q, we first compute its similarity to all memory chunks across each granularity level $g \in \{\text{session, turn, summary, keyword}\}$. Let $\boldsymbol{s}^g = [\text{sim}(q, M_1^g), \ldots, \text{sim}(q, M_n^g)]$ denote the similarity scores between q and n memories chunks at granularity g. We normalize $\boldsymbol{s}^g$ into a probability distribution $p^g(\boldsymbol{s}^g)$ and calculate its Shannon entropy:

$$H^g = -\sum_{i=1}^{n} p_i^g(\boldsymbol{s}^g) \log p_i^g(\boldsymbol{s}^g), \text{ where } p_i^g(\boldsymbol{s}^g) = \frac{\exp(\text{sim}(q, M_i^g)/\lambda)}{\sum_{j=1}^{n} \exp(\text{sim}(q, M_j^g)/\lambda)}, \forall i \in \{1, \ldots, n\}. \tag{3}$$

where H^g quantifies the uncertainty of matching q to memories at granularity g. λ is a hyperparameter to control the degree of entropy, and analyzed in Appendix D.

Soft Router Weights. Our motivation stems from that lower entropy H^g typically reflects higher confidence in precise matches (e.g., higher confidence indicates a clear correspondence between the query and its associated memory). Conversely, higher entropy suggests more ambiguous matches (e.g., lower confidence implies uncertainty about the query's corresponding memory). To capture this behavior, we assign weights to granularities by normalizing their inverse entropy:

$$w^g = \frac{1/H^g}{\sum_{g'=1}^{G} 1/H^{g'}}, \tag{4}$$

where G denotes the total number of granularities. This formulation ensures that granularities with lower entropy (indicating higher certainty) are assigned greater weights. Consequently, memories are reweighted to emphasize those associated with the query's most confident granularities, eliminating the need for manual intervention.

2.4 Memory Retrieval and Filter

After constructing the multi-granularity memory associations and determining granularity weights, we retrieve relevant memories for a query q by leveraging the graph-structured memory $\{M_i, A_i\} \in \mathcal{M} \times \mathcal{A}$. To fully utilize inter-memory relationships, we employ the Personalized PageRank (PPR) algorithm [10] for context-aware ranking. The initial relevance score for each memory node M_i is computed as a weighted aggregation across all granularities, using the router-assigned weights w^g from Equation 4:

$$\text{score}_i = \sum_{g=1}^{G} w^g \cdot \text{sim}(q, M_i^g), \tag{5}$$

where $\text{sim}(q, M_i^g)$ measures the cosine similarity between the query embedding $e(q)$ and the granularity-specific embedding $e(M_i^g)$. This score reflects the query's direct relevance to M_i across multiple granularities.

Using the initial scores score_i as personalized starting probabilities, we select the top α nodes as seed nodes and run the PPR algorithm on the memory association graph. This propagates relevance signals through the graph structure, emphasizing memories that are both directly relevant to the query and densely connected to other high-value memories. The final PPR scores rank memories by their global importance within the graph context. After convergence, we select the top-K nodes as candidate contexts by their final PPR scores. The impact of K is empirically analyzed in § 3.4.

LLM-Based Redundancy Filtering. To minimize noise and eliminate redundancy in the retrieved multi-granularity memories, we employ an LLM-based filtering mechanism. This mechanism processes the top-K memories alongside the query q, using a carefully designed prompt (see Appendix H) to identify and discard irrelevant or repetitive content. As a result, the final context provided to the response generator is refined to focus exclusively on the most critical information relevant to q, ensuring a more concise and personalized response. We present case study on multi-granularity information in Appendix I.2 and case study on filtering in Appendix I.3.

3 Experiments

3.1 Experimental Settings

Dataset and Metrics. Experiments are conducted on four comprehensive long-term memory datasets: LoCoMo-10 [30], Long-MT-Bench+ [33], LongMemEval-s [50], and LongMemEval-m [50], all focused on evaluating the capabilities of LLM agents in long-term conversations. Since our tasks require no training, the whole QA pairs in datasets are used as test set. Detailed dataset statistics are provided in Appendix A. To comprehensively evaluate model performance, we employ multiple metrics: F1 scores (as used by Maharana et al. [30]), BLEU, BERTScore, and ROUGE scores (following Pan et al. [33]). Additionally, we introduce GPT4o-as-Judge (GPT4o-J) [58], an evaluation setting where GPT4o assesses the alignment of a model's response with the reference answer. The evaluation prompts are provided in Appendix H.

Baselines. We compare MemGAS against various methods. (1) *Full History*: which utilizes all the latest conversation records, incorporating up to 128k tokens of context. **Two strong retrieval models**: (2) *MPNet* [43] and (3) *Contriever* [13]. **Three memory-based conversational models**: (4) *LLM-Rsum* [49], which uses LLMs to recursively summarize and update memory for contextually relevant responses; (5) *MPC* [19], which leverages pre-trained LLMs to create high-quality conversational agents; (6) *SeCom* [33], which segments memory into coherent topics and applies compression-based denoising to boost retrieval accuracy; **Two structured RAG models**: (7) *HippoRAG 2* [8], which integrates knowledge graphs for efficient retrieval, and (8) *RAPOTR* [42], which enhances retrieval via recursive summary and hierarchical clustering into a tree structure. We also provide a comparison of different retrievers in Appendix C and error analysis in Appendix G.

Implementation Details. For our experiments, we employ 'gpt-4o-mini-2024-07-18' as backbone for all tasks, including multi-granularity information generation and question-answer generation. To ensure fair comparisons, all baselines are implemented using consistent generation prompts. The temperature of LLMs is set to 0 to facilitate reproducibility. Additionally, a uniform top-3 document retrieval setting is applied across all models in the main results. We use Contriever [13] as the encoding model to generate embeddings for memory texts. The prompt templates are available in Appendix H. Detailed cost and efficiency are analyzed in Appendix F.

3.2 Overall Results

We present the results for Question Answering and Retrieval in Table 1 and Table 2, respectively. We also compare performence of single-granularity and multi-granularity in Table 3. Note that LongMTBench+ does not provide a ground-truth retrieval corpus and is therefore not included. RAPTOR cannot be evaluated for retrieval performance because its retrieved information cannot be judged. Due to high runtime of RAPROT and HippoRAG on LongMemEval-m, we do not obtain results on them. Below, we provide analysis of these results.

Table 1: QA Performance. Contriever is used as the retrieval backbone (except for Full History and MPNet), with GPT4o-mini as the generator, and the top 3 results are retrieved as input context. **Bolded** values indicate the best results, while underlined values represent the second-best results. All metrics are the larger, the better.

Model	GPT4o-J	F1	BLEU	Rouge1	Rouge2	RougeL	BERTScore
LongMemEval-s							
Full History	50.60	11.48	1.40	12.10	5.47	10.85	83.07
MPNet [43]	53.20	13.96	2.21	14.49	6.78	12.93	83.72
Contriever [13]	55.40	13.78	2.21	14.46	6.93	12.89	83.70
LLM-Rsum [49]	35.40	12.29	2.09	13.01	5.55	11.52	83.60
MPC [19]	53.80	13.60	1.74	14.27	6.49	12.95	83.49
SeCom [33]	56.00	12.95	2.25	13.80	6.09	11.93	83.51
HippoRAG 2[9]	57.60	14.73	2.15	15.30	7.36	13.83	83.86
RAPTOR [42]	32.20	12.08	1.90	12.73	5.82	11.25	83.50
MemGAS (Ours)	**60.20**	**20.38**	**4.22**	**21.05**	**10.47**	**19.47**	**85.21**
LongMemEval-m							
Full History	12.20	5.70	0.78	6.27	2.08	5.28	81.62
MPNet [43]	37.80	10.76	1.70	11.46	4.76	10.03	83.09
Contriever [13]	42.80	11.88	1.66	12.56	5.63	11.02	83.28
LLM-Rsum [49]	23.80	10.04	1.70	10.89	4.26	9.21	83.12
MPC [19]	37.80	11.28	1.37	11.93	5.12	10.57	82.98
SeCom [33]	42.80	11.33	1.79	12.03	5.07	10.49	83.16
MemGAS (Ours)	**45.40**	**16.85**	**3.39**	**17.60**	**8.25**	**16.14**	**84.69**
LOCOMO-10							
Full History	33.43	12.23	1.84	12.70	5.66	11.73	84.07
MPNet [43]	38.07	14.44	2.35	14.90	6.83	13.90	84.42
Contriever [13]	40.33	15.66	2.67	16.01	7.68	15.00	84.65
LLM-Rsum [49]	22.56	9.14	0.99	9.82	3.38	8.98	83.45
MPC [19]	40.38	14.81	1.99	15.10	6.83	14.13	84.43
SeCom [33]	44.21	13.79	2.30	14.28	6.17	13.30	84.04
HippoRAG 2[9]	**45.62**	16.66	2.91	17.01	8.27	15.93	84.88
RAPTOR [42]	31.72	14.55	2.88	15.09	7.49	14.18	84.48
MemGAS (Ours)	41.07	**17.66**	**3.61**	**18.00**	**8.93**	**16.99**	**85.13**
LongMTBench+							
Full History	67.44	36.07	11.32	37.90	20.51	29.25	87.81
MPNet [43]	63.89	36.09	11.26	38.39	20.58	28.86	87.84
Contriever [13]	63.54	36.30	11.59	38.17	21.65	29.67	87.90
LLM-Rsum [49]	24.65	26.58	6.91	29.23	11.93	20.90	86.11
MPC [19]	61.81	31.52	7.97	33.56	17.27	25.20	86.51
SeCom [33]	64.58	36.68	12.01	38.81	21.44	29.65	87.88
HippoRAG 2[9]	63.54	35.64	11.05	37.61	20.37	28.76	87.70
RAPTOR [42]	59.72	37.69	13.47	40.08	21.68	30.88	88.38
MemGAS (Ours)	**69.44**	**41.49**	**15.62**	**43.69**	**24.45**	**34.66**	**88.96**

Question Answering Results. As presented in Table 1, MemGAS consistently outperforms other methods across most datasets and evaluation metrics. Unlike the Full History approach, which introduces noise by utilizing all historical information, MemGAS excels by effectively consolidating and retrieving only the most relevant memories. Other baselines, such as LLM-Rsum and SeCom, although operating at specific granularities, lack the capability to integrate multi-granular associations

Table 2: Retrieval Performance. All methods are based on Contriever as the retriever, except for MPNet.

Model	Recall@3	NDCG@3	Recall@5	NDCG@5	Recall@10	NDCG@10
			LongMemEval-s			
MPNet [43]	66.17	75.47	76.38	78.29	85.11	80.63
Contriever [13]	71.06	79.72	81.28	82.47	90.00	84.29
LLM-Rsum [49]	67.23	78.33	79.79	81.76	87.66	83.28
MPC [19]	60.00	70.90	68.09	73.27	80.00	76.59
SeCom [33]	71.06	80.88	80.43	83.08	89.15	85.11
HippoRAG 2[9]	75.53	85.44	84.68	87.32	91.28	88.73
MemGAS (Ours)	**78.51**	**86.83**	**88.94**	**88.77**	**94.47**	**89.96**
			LongMemEval-m			
MPNet [43]	37.02	48.68	45.74	52.51	61.28	56.55
Contriever [13]	44.26	56.11	53.40	59.44	65.96	62.74
LLM-Rsum [49]	23.19	32.10	31.70	36.99	43.62	41.40
MPC [19]	35.96	47.51	42.55	50.36	54.26	53.88
SeCom [33]	44.26	56.61	55.32	60.76	66.60	63.78
MemGAS (Ours)	**51.06**	**61.36**	**63.62**	**66.07**	**77.02**	**69.46**
			LOCOMO-10			
MPNet [43]	45.92	47.71	53.98	51.79	68.58	56.88
Contriever [13]	49.90	52.15	58.26	56.29	71.80	60.92
LLM-Rsum [49]	47.23	48.99	59.01	54.58	74.97	60.07
MPC [19]	49.50	51.47	57.45	55.53	71.85	60.47
SeCom [33]	55.24	57.90	64.80	62.36	78.30	66.97
HippoRAG 2[9]	56.60	58.37	65.06	62.50	78.05	66.79
MemGAS (Ours)	**57.30**	**58.76**	**67.32**	**63.62**	**81.82**	**68.42**

Table 3: Single-Granularity vs Multi-Granularity on QA performance, which is evaluated using top 3 retrieved results.

Granularity	**Method**	**QA Performance**			**Retrieval Performance**		
		GPT4o-J	F1	RogueL	Recall@3	Recall@5	Recall@10
Single-Granularity	Session-level	55.40	13.78	12.89	71.06	81.28	90.00
	Turn-level	57.60	14.94	14.12	73.62	84.68	91.91
	Summary-level	28.80	10.66	9.96	70.43	80.00	88.30
	Keyword-level	17.20	9.05	8.20	62.98	74.68	82.34
Multi-Granularity	Combination	56.60	14.59	13.70	76.17	87.23	91.91
	MemGAS	**60.20**	**20.38**	**19.47**	**78.51**	**88.94**	**94.47**

between memory, resulting in suboptimal outcomes. Besides, methods like HippoRAG 2 and RAPTOR, while establishing connections between memory units, fail to construct multi-granular relationships and selection mechanisms effectively, limiting their performance. MemGAS performs better through its dynamic construction and adaptive router of multi-granular memory units and redundancy filtering, emphasizing the critical role of association and selection in effective long-term memory management.

Retrieval Results. As shown in Table 2, our MemGAS demonstrates outstanding performance across all datasets, consistently achieving the highest Recall and NDCG metrics. These results highlight the effectiveness and robustness of our approach, addressing key challenges in long-term memory construction and retrieval, and ensuring that queries are matched with the most relevant context.

Single-granularity vs Multi-granularity Results. We compare our MemGAS with Single-granularity settings including (session-level, turn-level, summary-level and keyword-level), and Multi-granularity by directly combining text. Table 3 shows that single-granularity settings underperform compared to multi-granularity approaches in both QA and retrieval tasks. This demonstrates the

Table 4: Ablation Study on the LongMemeval-s Dataset. QA performance is evaluated using top 3 retrieved results. The filtering setting is applied as a post-retrieval process. Thus, retrieval results are not included.

Method	QA Performance			Retrieval Performance		
	GPT4o-J	F1	RogueL	Recall@3	Recall@5	Recall@10
MemGAS	60.20	**20.38**	**19.47**	**78.51**	**88.94**	**94.47**
w/o Gaussian Mixture Model	57.20	19.49	18.68	76.38	85.53	91.28
w/o Personalized PageRank	56.60	19.76	18.85	75.96	85.96	90.64
w/o Memory Association	56.80	17.69	19.00	74.89	85.74	91.49
w/o Soft Router	56.60	18.88	18.62	75.53	85.74	92.34
w/o Filtering	**60.60**	14.37	13.54	-	-	-
w/o All	55.40	13.78	12.89	71.06	81.28	90.00

effectiveness of integrating multi-granular information in the long-term memory of agents. Furthermore, MemGAS achieves the best results, surpassing the simple combination of multi-granularity text. This indicates that the proposed memory association and entropy-based router modules effectively utilize multi-granular information, leading to significant performance gains. Comprehensive analyses across datasets and metrics further support these findings (See in Appendix B).

3.3 Ablation Study

The ablation study in Table 4 illustrates the contributions of various components to the QA and retrieval performance of MemGAS. In the w/o Gaussian Mixture Model setting, the system links using only the most similar node, resulting in a noticeable performance drop. Similarly, the w/o Personalized PageRank setting, which uses one-hot nodes retrieval, leads to reduced recall and QA scores, emphasizing the importance of multi-hop retrieval. Memory Association and Router also contribute significantly, with their absence leading to declines in QA and retrieval results. Interestingly, removing filtering boosts GPT4o-J's QA performance. It can be explained that retaining more information aligns better with the LLM judge's preference for comprehensive responses. The most significant drop occurs when all components are removed, with the F1 score decreasing from 20.38 to 13.78, highlighting their combined importance.

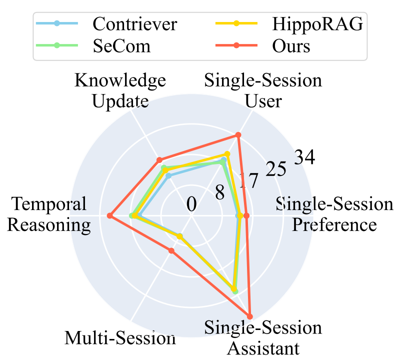

Figure 3: Comparison of F1 Across Different Query Types in LongMemEval-s.

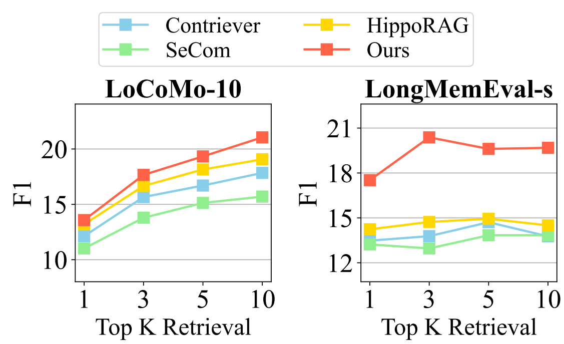

Figure 4: Comparison on Different Top-K Retrieval.

3.4 Detailed Comparison Analysis

In this section, we present a comprehensive analysis comparing our approach with baselines across different query types and different Top-K retrieval settings.

Comparison on different query types. The results in Figure 6 highlight the performance across different query types. Our MemGAS consistently demonstrates superior performance, particularly excelling in *multi-hop retrieval* on Locomo-10 and *multi-session* on LongMemEval-s. This suggests that the memory association mechanism in MemGAS effectively identifies highly relevant sessions, enabling enhanced multi-hop reasoning and better integration of knowledge. Additionally, our method performs well across single-session, temporal reasoning, and knowledge update, demonstrating its strength in addressing both simple and complex query types. We also provide a case study for different query types in Appendix I.1, and comparison with other datasets and metrics in Appendix E.

Comparison on different Top-K retrieval. The results shown in Figure 4 demonstrate the performance of various Top-K retrieval settings. On the LoCoMo-10 and LongMemEval-s datasets, our model consistently surpasses baseline methods in F1 scores as Top-K increases from 1 to 10. This trend highlights that retrieving more relevant context significantly enhances accuracy. Note that on LongMemEval-s dataset, while F1 scores initially improve at lower Top-K values, performance declines at higher Top-K levels, suggesting that the longer context introduces noise, which negatively impacts the model's effectiveness.

4 Related Work

Retrieval Methods. Recent advancements in retrieval methods have greatly enhanced information retrieval performance[40, 43, 41]. BM25 [40], a classic sparse retrieval method, uses a probabilistic model to improve query relevance. In contrast, dense retrieval methods excel at capturing semantic similarity. For instance, DPR [16] employs dual-encoder models to encode queries and documents into dense vectors for efficient similarity search. Similarly, Contriever [13] leverages contrastive learning to enhance semantic understanding. Advanced methods like E5 [48], BGE [29], and GTE [24] further utilize pre-trained or fine-tuned transformers for robust and efficient semantic retrieval.

Long Term Memory Management. With the development of LLMs, the user-assistant conversation becomes longer and contains various topics [28, 5, 37, 32, 55, 56, 51, 5, 18], which introduces challenges for preserving the user's long-term memory. Management in long-term memory often involves segmentation [33], summary [17], compression [3], forgetting and updating [1, 49, 59]. For example, several approaches [49, 42, 45, 26, 28] focus on generating memory summary as records to enable more accurate retrieval. Besides, some methods [33, 20, 52, 2] leverage compression techniques to reduce memory size while preserving essential information. The forgetting mechanisms [59, 14] address the need to remove obsolete or irrelevant memories while maintaining model performance. Some integrated methods [44, 31, 21] combine memory retention, update to enable personalized and contextually relevant long-term dialogue. However, Existing approaches typically focus on single-granularity segmentation strategies, such as session/turn, to organize and manage long-term memory. Whereas our work leverages multi-granular information for better adaptive memory selection.

Structural Memory Management. Additionally, existing works employ structured paradigms for memory or knowledge base organization [4, 25, 55, 56, 51, 11, 57, 53]. HippoRAG [8, 9] builds entity-centric knowledge graphs inspired by hippocampal indexing theory [46], while Graph-CoT [15] and G-Retriever [12] integrate graph reasoning for interactive retrieval or generation, whereas LightRAG [7] and GraphRAG [6] optimize retrieval and summary via graph structures. MemTree and RAPTOR [39, 42] utilize recursive embedding, clustering, and summarization of text chunks to construct a hierarchical tree, while StructRAG [25] enhances reasoning by leveraging multiple structured formats. While existing methods focus on single-granularity memory modeling, lacking cross-granularity interactions. In contrast, our MemGAS, which leverages multi-granularity association and adaptive selection to construct consolidated memory structures and optimize retrieval efficiency.

5 Conclusion

In this paper, we proposed MemGAS, a novel framework for long-term memory construction and retrieval that integrates multi-granular memory units and enables adaptive selection and retrieval. By leveraging human-inspired memory mechanisms through Gaussian Mixture Models and an entropy-based multi-granularity router, MemGAS effectively addresses challenges of memory connection and selection. Experimental results across four benchmarks demonstrate that MemGAS significantly outperforms state-of-the-art baselines in both QA and retrieval tasks, highlighting its robustness and superiority in managing long-term memory for conversational agents.

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A Datasets Statistics.

Table 5: Dataset statistics. The term 'Avg.' (e.g., Avg. Sessions) represents the average number corresponding to each conversation.

Dataset	LoCoMo-10	Long-MT-Bench+	LongMemEval-s	LongMemEval-m
Total Conversations	10	11	500	500
Avg. Sessions	27.2	4.9	50.2	501.9
Avg. Query	198.6	26.2	1.0	1.0
Avg. Token	20,078.9	19,194.8	103,137.4	1,019,116.7
Sessions Dates	✓	✗	✓	✓
Retrieval Ground-Truth	✓	✗	✓	✓
QA Ground-Truth	✓	✓	✓	✓
Conversation Subject	User-User	User-AI	User-AI	User-AI

We evaluate the proposed method with the following benchmarks:

- **LongMemEval-s** [50] and **LongMemEval-m** [50] datasets are benchmarks designed to evaluate long-term memory in chat assistants. LongMemEval-s involves 50 sessions per question with an average of 115k tokens, making it a compact yet challenging dataset. LongMemEval-m, on the other hand, spans 500 sessions per question, resulting in avg. 1.5 million tokens pre conversations, offering a more extensive evaluation setting. Both datasets require AI systems to handle user-AI dialogues, dynamic memorization, and historical consistency.
- **LoCoMo** [30] dataset evaluates long-term memory in AI through lengthy conversations, averaging 300 turns, 9,000 tokens, and up to 35 sessions. LoCoMo-10, a publicly available subset of LoCoMo, includes 10 high-quality, long conversations with 27.2 sessions and 20,000 tokens on average. The goal of LoCoMo is to evaluate long-context LLMs and RAG systems, which, while improving memory performance, still fall short of human capabilities—particularly in temporal reasoning—underscoring the challenges of understanding long-range dependencies.
- **Long-MT-Bench+** [33] dataset is reconstructed from MT-Bench+ [28] by incorporating long-range questions to address the limited QA pairs and short dialogues. It merges five consecutive sessions into a single long-term conversation, improving its suitability for evaluating memory mechanisms. The dataset features 11 conversations with an average of 4.9 sessions and 19194.8 tokens. Unlike LoCoMo-10, it focuses on user-AI interactions, excluding session dates and retrieval ground truth.

B Comprehensive analysis of Single-Granularity and Multi-Granularity

B.1 QA Performance

The results in the Table 6 demonstrate the significant advantages of multi-granularity methods over single-granularity approaches. Across all datasets, multi-granularity methods outperform single-granularity methods in key evaluation metrics such as F1, BLEU, ROUGE, and BERTScore. For example, in the LongMemEval-m dataset, the best-performing single-granularity method (Turn-level) achieves an F1 score of 12.26, whereas the multi-granularity method achieves significantly higher F1 scores of 12.51 and 16.85. Similarly, in the LongMTBench+ dataset, the multi-granularity method achieves F1 scores of 37.16 and 41.49, outperforming the best single-granularity method, which scores 36.67. These results highlight the effectiveness of multi-granularity approaches in integrating information from different granularities, providing a more comprehensive understanding and significantly improving QA task performance.

Moreover, the proposed method, MemGAS reveals its superiority over simple multi-granularity combination methods. Across all datasets, MemGAS consistently achieves better performance not only in F1 scores but also across other evaluation metrics. For instance, in the LongMemEval-s dataset, MemGAS achieves an F1 score of 20.38, surpassing the combination method's score of 14.59. Similarly, in the LongMTBench+ dataset, MemGAS achieves an F1 score of 41.49, compared to 37.16 for the combination method. These results demonstrate that MemGAS leverages a more sophisticated

Table 6: QA performance.

Granularity	GPT4o-J	F1	BLEU	Rouge1	Rouge2	RougeL	BERTScore
LongMemEval-s							
Single-Granularity							
Session-level	55.40	13.78	2.21	14.46	6.93	12.89	83.70
Turn-level	<u>57.60</u>	<u>14.94</u>	<u>2.50</u>	<u>15.53</u>	<u>7.50</u>	<u>14.12</u>	83.93
Summary-level	28.80	10.66	1.78	11.51	4.61	9.96	83.26
Keyword-level	17.20	9.05	1.61	9.82	3.61	8.20	82.84
Multi-Granularity							
Combination	56.60	14.59	2.40	15.19	7.22	13.70	<u>83.96</u>
MemGAS	**60.20**	**20.38**	**4.22**	**21.05**	**10.47**	**19.47**	**85.21**
LongMemEval-m							
Single-Granularity							
Session-level	42.80	11.88	1.66	12.56	5.63	11.02	83.28
Turn-level	42.60	12.26	1.79	12.93	5.67	11.45	83.43
Summary-level	24.40	10.11	1.68	11.02	4.16	9.41	83.17
Keyword-level	15.00	8.29	1.54	9.16	3.26	7.43	82.72
Multi-Granularity							
Combination	**46.40**	<u>12.51</u>	<u>1.96</u>	<u>13.08</u>	<u>6.13</u>	<u>11.70</u>	<u>83.49</u>
MemGAS	<u>45.40</u>	**16.85**	**3.39**	**17.60**	**8.25**	**16.14**	**84.69**
LOCOMO-10							
Single-Granularity							
Session-level	40.33	15.66	2.67	16.01	7.68	15.00	84.65
Turn-level	<u>42.09</u>	<u>16.10</u>	<u>2.80</u>	<u>16.42</u>	<u>8.13</u>	<u>15.36</u>	<u>84.70</u>
Summary-level	19.08	9.78	0.83	10.38	3.28	9.41	83.69
Keyword-level	14.85	7.74	0.64	8.40	2.20	7.62	83.28
Multi-Granularity							
Combination	**42.80**	15.93	2.60	16.27	7.76	15.26	84.69
MemGAS	40.08	**17.66**	**3.61**	**18.00**	**8.93**	**16.99**	**85.13**
LongMTBench+							
Single-Granularity							
Session-level	63.54	36.30	11.59	38.17	21.65	29.67	87.90
Turn-level	67.01	36.67	<u>11.91</u>	39.06	20.88	30.13	<u>88.04</u>
Summary-level	21.18	28.30	8.37	31.12	13.11	22.55	86.74
Keyword-level	20.83	28.05	8.90	30.92	12.80	22.39	86.78
Multi-Granularity							
Combination	<u>67.01</u>	<u>37.16</u>	11.58	<u>39.11</u>	<u>21.95</u>	<u>30.24</u>	88.00
MemGAS	**69.44**	**41.49**	**15.62**	**43.69**	**24.45**	**34.66**	**88.96**

mechanism to capture the relationships and complementarities among different granularities. This enables MemGAS to achieve greater robustness and generalization, making it particularly effective for complex QA tasks.

B.2 Retrieval Performance

The results in Table 7 clearly demonstrate the advantages of the multi-granularity approach in retrieval tasks compared to single-granularity methods. Across all datasets, multi-granularity methods consistently achieve higher scores in key retrieval metrics such as Recall and NDCG. For example, in the LongMemEval-s dataset, the best single-granularity method (Turn-level) achieves Recall@10 of 91.91 and NDCG@10 of 86.73. However, the simplest multi-granularity combination approach further improves these scores to 91.91 and 88.03, respectively, demonstrating its superior retrieval effectiveness. Similarly, in the LongMemEval-m dataset, the Recall@10 and NDCG@10 of the multi-granularity approach reached at least 73.83 and 68.66, respectively, which significantly exceeded

Table 7: Retrieval Performance.

Granularity	Recall@3	NDCG@3	Recall@5	NDCG@5	Recall@10	NDCG@10
LongMemEval-s						
Single-Granularity						
Session-level	71.06	79.72	81.28	82.47	90.00	84.29
Turn-level	73.62	82.47	84.68	84.94	91.91	86.73
Summary-level	70.43	80.53	80.00	82.38	88.30	84.28
Keyword-level	62.98	74.69	74.68	77.76	82.34	79.50
Multi-Granularity						
Combination	76.17	84.68	87.23	87.00	91.91	88.03
MemGAS	**78.51**	**86.83**	**88.94**	**88.77**	**94.47**	**89.96**
LongMemEval-m						
Single-Granularity						
Session-level	44.26	56.11	53.40	59.44	65.96	62.74
Turn-level	44.68	55.06	57.66	60.03	70.21	63.43
Summary-level	41.06	54.68	52.34	58.64	65.53	62.31
Keyword-level	35.11	48.22	43.62	52.02	57.87	55.75
Multi-Granularity						
Combination	50.85	**61.50**	60.00	64.82	73.83	68.66
MemGAS	**51.06**	61.36	**63.62**	**66.07**	**77.02**	**69.46**
LOCOMO-10						
Single-Granularity						
Session-level	49.90	52.15	58.26	56.29	71.80	60.92
Turn-level	52.77	54.09	64.30	59.64	80.31	65.07
Summary-level	47.89	48.96	58.21	53.94	74.12	59.61
Keyword-level	29.05	29.72	40.33	35.46	65.11	44.11
Multi-Granularity						
Combination	53.98	55.89	63.80	60.61	78.70	65.64
MemGAS	**57.45**	**58.84**	**67.12**	**63.60**	**81.07**	**68.24**

the best results of the single-granularity approach (Recall@10 = 70.21 and NDCG@10 = 63.43, Turn-level). These findings underline the strength of multi-granularity approaches in leveraging diverse levels of information to retrieve more relevant results and improve overall retrieval quality.

The proposed method, MemGAS further demonstrates its superiority over simple multi-granularity combination methods. In all datasets, MemGAS consistently achieves better results. For example, in the LongMemEval-s dataset, MemGAS achieves Recall@10 of 94.47 and NDCG@10 of 89.96, surpassing the combination method's scores of 91.91 and 88.03. Similarly, in the LOCOMO-10 dataset, MemGAS achieves Recall@10 of 81.07 and NDCG@10 of 68.24, outperforming the combination method's scores of 78.70 and 65.64. These improvements demonstrate that MemGAS is not merely aggregating information from multiple granularities but instead employs a more advanced mechanism to capture the interdependencies and complementarities among different granularities. This enables MemGAS to deliver robust and versatile performance, making it highly effective for complex retrieval tasks that require the integration of multi-granularity information.

C Comparison based on Different Retriever

The results in the Table 8 demonstrate that MemGAS consistently outperforms all other methods across all metrics when evaluated with different base retrievers (MiniLM, MPNet, and Contriever) on the LoCoMo-10 dataset. For the MiniLM base retriever, MemGAS achieves the highest scores in all metrices, outperforming MiniLM, MPC, LLM-Rsum, and SeCom. This indicates that MemGAS is highly effective in leveraging the MiniLM retriever to improve retrieval performance.

When using the MPNet and Contriever base retrievers, MemGAS maintains its superiority across all metrics. For MPNet, MemGAS achieves a Recall@10 of 80.51 and NDCG@10 of 65.48, which

Table 8: Retrieval Performance based on different retriever on LoCoMo-10. 'facebook/contriever', 'sentence-transformers/multi-qa-mpnet-base-cos-v1', 'sentence-transformers/multi-qa-MiniLM-L6-cos-v1'

Model	**Recall@3**	**NDCG@3**	**Recall@5**	**NDCG@5**	**Recall@10**	**NDCG@10**
Base Retriever: MiniLM						
MiniLM [43]	42.55	44.19	52.37	49.01	67.98	54.59
MPC [19]	42.30	43.59	51.21	48.08	68.03	54.03
LLM-Rsum [49]	44.76	46.82	54.73	51.64	<u>72.16</u>	<u>57.52</u>
SeCom [33]	<u>45.77</u>	<u>47.25</u>	<u>54.93</u>	<u>51.70</u>	71.15	57.37
MemGAS	**47.73**	**49.11**	**56.60**	**53.46**	**71.30**	**58.59**
Base Retriever: MPNet						
MPNet [43]	45.92	47.71	53.98	51.79	68.58	56.88
MPC [19]	45.47	47.35	54.08	51.68	68.28	56.59
LLM-Rsum [49]	<u>49.50</u>	<u>51.15</u>	<u>59.47</u>	<u>56.16</u>	<u>76.64</u>	<u>61.99</u>
SeCom [33]	47.53	49.03	57.05	53.57	70.90	58.57
MemGAS	**52.77**	**54.63**	**62.79**	**59.56**	**80.51**	**65.48**
Base Retriever: Contriever						
Contriever [13]	49.90	52.15	58.26	56.29	71.80	60.92
MPC [19]	49.50	51.47	57.45	55.53	71.85	60.47
LLM-Rsum [49]	47.23	48.99	59.01	54.58	74.97	60.07
SeCom [33]	<u>55.24</u>	<u>57.90</u>	<u>64.80</u>	<u>62.36</u>	<u>78.30</u>	<u>66.97</u>
MemGAS	**57.30**	**58.76**	**67.32**	**63.62**	**81.82**	**68.42**

are significantly higher than the other methods. Similarly, for the Contriever retriever, MemGAS obtains the best Recall@10 (81.82) and NDCG@10 (68.42). These results highlight the robustness and adaptability of MemGAS demonstrating its ability to outperform alternative methods regardless of the underlying retriever model.

D Hyperparameter Sensitivity Analysis

We evaluate the hyperparameter entropy degree λ in Equation 3 and the number of seed nodes α in Equation 5. The results of the hyperparameter α in Figure 5 show that performance metrics, including nDCG and Recall, improve as α increases, reaching their peak when α is set to a moderate value around 15. Beyond this point, performance declines, indicating that a balanced setting of α is crucial for optimal results. When α is too small, the model tends to underfit, struggling to capture sufficient seed nodes in the data. On the other hand, excessively large values of α cause the model to lose the ability to explore the memory graph. When α is set to 15, the model often achieves better performance across various metrics.

For the hyperparameter entropy degree λ, a similar trend is observed in Figure 5. Smaller values of λ lead to steady improvements in performance, with the metrics reaching their highest levels when λ is set to a value slightly above the minimum, around 0.2. However, as λ increases further, performance begins to degrade, emphasizing the importance. Extremely small values may fail to leverage the trade-offs controlled by λ, while larger values lead to the same entropy for different granularities. When λ is kept around 0.2, the model often delivers superior performance.

E Comparison on different query types

The radar charts in Figure 6 demonstrate the performance of different methods across diverse query dimensions. Our approach consistently outperforms baseline methods, particularly excelling in *multi-hop retrieval* on LoCoMo-10 (Figures 6a, 6b) and *multi-session* on LongMemEval-s (Figures 6c, 6d). This highlights the effectiveness of our framework in addressing both complex reasoning tasks and simpler retrieval-based queries. In *multi-hop retrieval* task, our method achieves notably

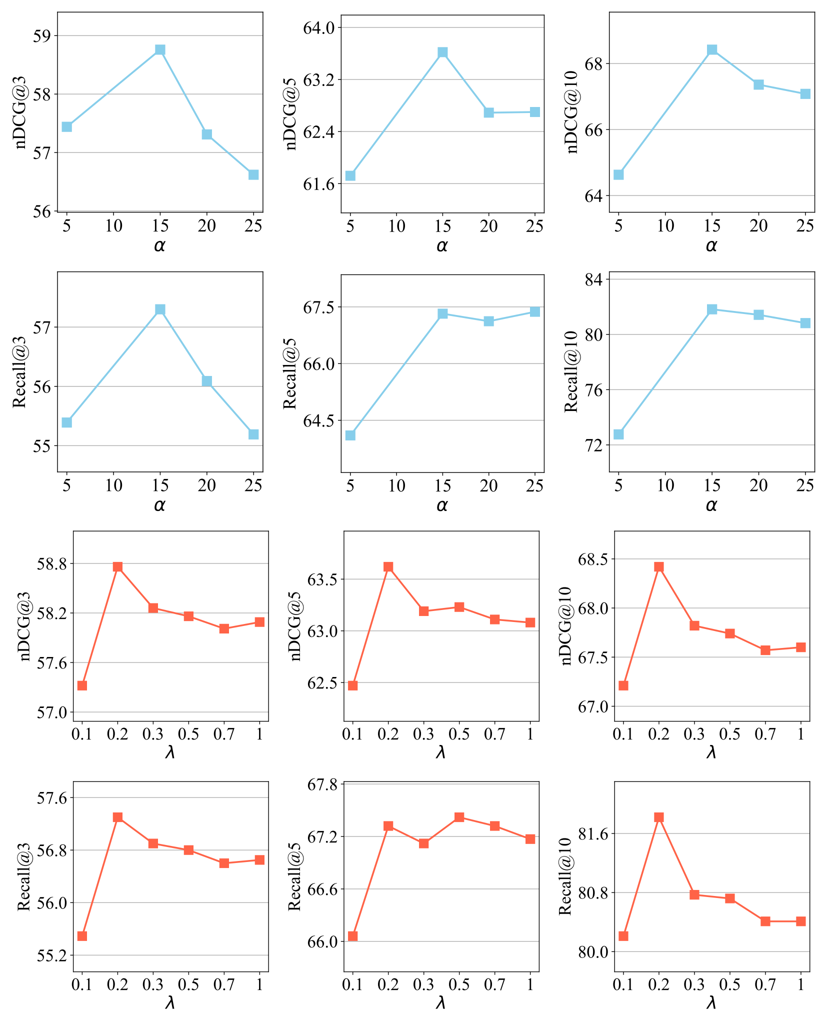


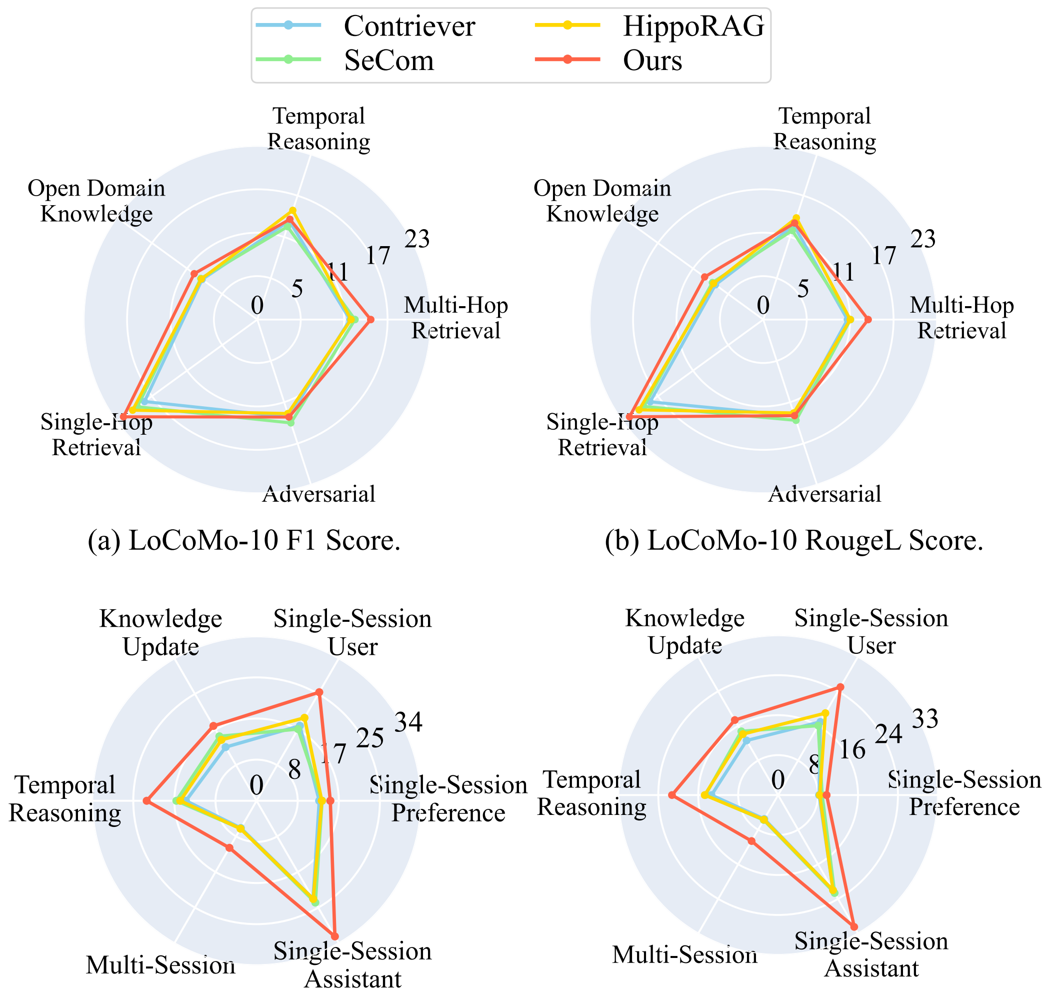
Figure 5: Hyperparameters Analysis on Locomo-10 dataset.

Table 9: Comparing the total input and output token consumption of baselines on the LongMemEval-s dataset. The LongMemEval-s dataset contains approximately 51.6M corpus tokens. 'M' means million.

	MemGAS	HippoRAG 2	RAPTOR	SeCom	LLM-Rsum
Input Tokens	52.9M (100 %)	111.1M (210.0 %)	62.6M (118.3 %)	106.2M (200 %)	58.3M (110 %)
Output Tokens	5.2 M (100 %)	10.9M (209.6 %)	0.73M (14.0 %)	71.1M (1367 %)	16.3M (313 %)

higher F1 and RougeL scores, showcasing its ability to retrieve and integrate information across multiple steps. Similarly, in *multi-session* task, the model effectively captures relevant historical sessions, enabling context-aware and accurate responses. These strengths are driven by the memory association mechanisms that enhance reasoning and knowledge integration.

Our framework also demonstrates strong performance in *temporal reasoning*, *knowledge update*, and *single-session* tasks, as shown by consistently higher scores across these axes. This indicates its ability to adapt to dynamic information and maintain relevance in evolving contexts. Furthermore, the model performs well in adversarial and open-domain knowledge tasks, reflecting its robustness and versatility. The results emphasize the comprehensive improvements achieved by our method across a wide range of query types, showcasing its capability to handle both simple and complex scenarios effectively. These findings underline the model's adaptability and suitability for real-world applications requiring diverse and dynamic query handling.



(a) LoCoMo-10 F1 Score.

(b) LoCoMo-10 RougeL Score.

(c) LongMemEval-s F1 Score.

(d) LongMemEval-s RougeL Score.

Figure 6: Comparing the F1 and RougeL socre of different query types on LoCoMo-10 and LongMemEval-s datasets.

F Cost and Efficiency Analysis

The results in Table 9 clearly demonstrate the exceptional efficiency of MemGAS compared to other baselines on the LongMemEval-s dataset. MemGAS processes only 52.9 M input tokens, which matches the total corpus size, while all other methods require significantly more tokens. For example, HippoRAG 2 processes over 111.1 M input tokens, RAPTOR uses over 62.6 M, and even the relatively efficient LLM-Rsum exceeds MemGAS . Similarly, in terms of output token usage, MemGAS generates only 5.2 M tokens, far fewer than the massive outputs of HippoRAG 2 and SeCom, which produce over 100 M and 70 M tokens, respectively. MemGAS achieves this remarkable efficiency without compromising performance, making it a highly effective and scalable solution for handling large-scale datasets. This demonstrates MemGAS 's ability to minimize computational costs while maintaining state-of-the-art results.

G Error Analysis

The error analysis presented in Figure 7 highlights the performance across three datasets: LoCoMo-10, LongMemEval-m, and LongMemEval-s. A key observation is that a significant portion of queries in the LongMemEval datasets lack correct retrieval results, as shown by the high percentage of "Wrong

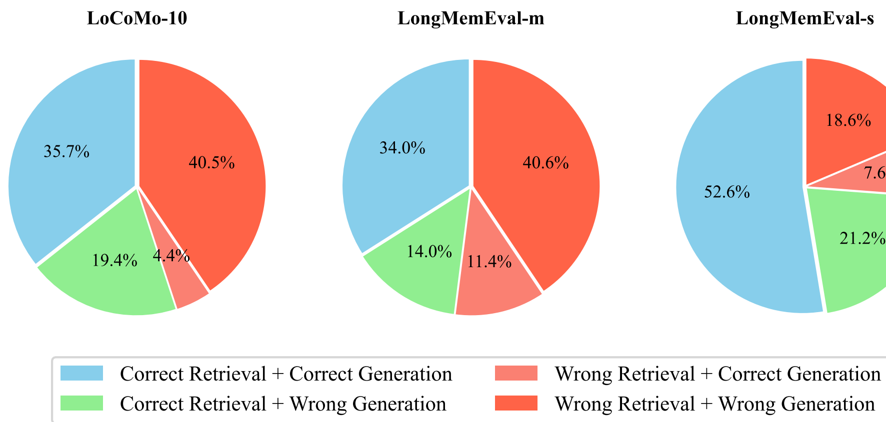


Figure 7: Error Analysis.

Retrieval + Wrong Generation" (40.6% in LongMemEval-m and 18.6% in LongMemEval-s). Our method effectively identifies cases where the retrieval corpus does not contain information related to the query, responding with "you don't mention the related information." This approach reduces hallucinations caused by large language models (LLMs) being overly confident and generating fabricated content. For example, in LongMemEval-s, "Correct Retrieval + Correct Generation" accounts for 52.6%, demonstrating the method's reliability in handling scenarios with relevant information.

H LLM Prompts

Figures 8–11 present the key prompt templates used in our multi-stage processing pipeline. Figure 8 defines prompts for multi-granularity information generation, where the model is asked to produce both high-level summaries and fine-grained keyword lists from user-assistant dialogue histories. Figure 9 introduces a filtering prompt that selects only the content relevant to the input question, operating over the structured outputs (session, summary, and keywords) while preserving original token fidelity. Figure 10 provides the instruction for generating final responses, guiding the model to produce concise and coherent answers grounded in the filtered history. Finally, Figure 11 shows the evaluation prompt used to assess answer correctness, following the setup of prior work [50, 58]. This prompt asks GPT-4o to compare model outputs against reference answers and return a binary decision without paraphrasing. Together, these prompts enable modular abstraction, filtering, reasoning, and automatic evaluation across the dialogue memory pipeline.

Prompt for Multi-granularity Information Generation

Summary Generation:

```
Below is an user-AI assistant dialogue memory.  Please summarize the
following dialogue as concisely as possible in a short paragraph, extracting
the main themes and key information.
```

Keyword Generation:

```
Below is an user-AI assistant dialogue memory.  Please extract the most
relevant keywords, separated by semicolon.
```

Figure 8: Prompt for Multi-granularity Information Generation.

I Case Study

This section presents a series of case studies comparing QA results with other methods, highlighting the model's capabilities in handling multi-granularity information generation and filtering. Through

Prompt for Multi-granularity Information Filter

```
You are an intelligent dialog bot. You will be shown History Dialogs and
corresponding multi-granular information. Filter the History Dialogs,
summaries, and keywords to extract only the parts directly relevant to the
Question. Preserve original tokens, do not paraphrase. Remove irrelevant
turns, redundant info, and non-essential details.
History Dialogs: {retrieved_texts}
Question Date: {question_date}
Question: {question}
Answer:
```

Figure 9: Prompt for Multi-granularity Information Filter.

Prompt for QA

```
You are an intelligent dialog bot. You will be shown History Dialogs.
Please read, memorize, and understand the given Dialogs, then generate one
concise, coherent and helpful response for the Question.
History Dialogs: { retrieved_texts}
Question Date: {question_date}
Question: {question}
```

Figure 10: Prompt for QA, which follows Lu et al. [28], Pan et al. [33]

Prompt for GPT4o-as-Judge

```
I will give you a question, a reference answer, and a response from a
model. Please answer [[yes]] if the response contains the reference answer.
Otherwise, answer [[no]]. If the response is equivalent to the correct
answer or contains all the intermediate steps to get the reference answer,
you should also answer [[yes]]. If the response only contains a subset of
the information required by the answer, answer [[no]].
[User Question]
question
[The Start of Reference Answer]
answer
[The End of Reference Answer]
[The Start of Model's Response]
response
[The End of Model's Response]
Is the model response correct? Answer [[yes]] or [[no]] only.
```

Figure 11: Prompt for GPT4o-as-Judge (Single), which follows Wu et al. [50], Zheng et al. [58]

these examples, we demonstrate how the model excels in maintaining factual consistency, extracting structured information, and filtering query-relevant responses effectively.

I.1 Case Study on QA Comparison

Figure 12 presents a comparative case study of three representative questions requiring multi-session aggregation or temporal reasoning. Across all cases, MemGAS consistently provides responses that align with the ground-truth answers, while baseline methods exhibit various limitations. In the first example, related to course completion, only MemGAS correctly integrates dispersed session information to produce the total number of courses. Other methods either rely on partial evidence or explicitly request further clarification. In the second case, which involves counting tomato and cucumber plants, most baselines retrieve incomplete or vague information, whereas MemGAS retrieves and combines both quantities explicitly. In the final temporal reasoning case, MemGAS successfully grounds the referenced date and computes the correct number of days, while others fail to locate the relevant temporal anchor or return fallback prompts. These results indicate that MemGAS is better equipped to support multi-turn factual consistency and basic temporal inference within multi-session contexts.

I.2 Case Study on Multi-granularity Information Generation

Figures 13 and 14 illustrate the model's ability to generate multi-granularity information from multi-turn user sessions. In both cases, the model extracts structured summaries and relevant keywords that reflect different levels of semantic abstraction. In Figure 13, the model distills a task-oriented request regarding energy-efficient industrial equipment into a concise summary, while preserving key product attributes and application contexts. The corresponding keywords focus on sales-related and functional concepts such as "pressure tanks" and "energy saving." In contrast, Figure 14 captures a more narrative and emotive user intent centered around personal heritage and item preservation. The summary highlights the user's goal of reclaiming and documenting a family-owned antique, while the extracted keywords reflect fine-grained care instructions and conservation practices. These examples demonstrate the model's capability to abstract dialogue content at varying semantic resolutions, enabling downstream applications such as memory retrieval, personalized assistant planning, and contextual reasoning. The structured outputs also suggest potential for use in grounding generation or summary-based retrieval settings.

I.3 Case Study on Multi-granularity Filter

Figures 15 and 16 demonstrate the effectiveness of multi-granularity response filtering in supporting user queries grounded in prior conversational context. In both cases, the user's query refers implicitly to entities previously mentioned across earlier sessions. The system retrieves multiple memory candidates and decomposes their content into structured representations at three granularity levels: session, summary, and keyword. This layered representation facilitates accurate entity matching and selective response generation. In Figure 15, although both memory candidates are sustainability-related, only one contains a precise mention of "Patagonia," which is correctly surfaced in the final response. Similarly, Figure 16 involves a comparative query referencing companies aligned with Triumvirate's values. The system filters through related prior mentions and correctly extracts "Patagonia" and "Southwest Airlines" from the relevant context. Notably, unrelated memories (e.g., on kitchen sustainability) are excluded from influencing the final answer. These examples highlight how multi-level content decomposition improves discourse grounding and allows retrieval systems to move beyond surface keyword overlap, ensuring the returned response aligns with both the semantic focus and factual detail required by the user query.

J Limitations

While our proposed framework demonstrates significant advancements in addressing multi-granularity memory association and adaptive granularity selection, it is not without limitations. There are some key areas where MemGAS can be further improved. Our method depends on LLM-generated summaries and keywords. The quality of memory construction in MemGAS heavily relies on the ability of LLMs to generate accurate and relevant summaries and keywords. Errors or biases in these generated outputs could propagate through the memory system, affecting the accuracy of memory association and retrieval. The scalability of large-scale memory systems needs to be improved. Although MemGAS leverages techniques such as Gaussian Mixture Models and Personalized PageRank to manage and retrieve memory, its computational efficiency may degrade as the memory size grows substantially. This could limit its applicability in scenarios requiring the processing of extensive conversational histories or highly dynamic memory updates.

Case of compare multi-method response

Case 1: Multi-session Question – Online Course Completion
Query: How many online courses have I completed in total?
Ground-Truth Answer: 5
Responses of Different Methods:

- **MemGAS** : You have completed a total of five online courses. ✓
- **SeCom**: Based on the dialogues, you have completed two courses on edX. However, there are no mentions of any additional courses completed, so if those are the only ones you've undertaken, your total number of completed online courses is **two**. ✗
- **MPC**: You have completed a total of 2 online courses on edX. After finishing the Stanford NLP with Deep Learning Specialization on Coursera, which consists of 5 courses, you will have completed 7 courses. As of now, you have completed **2** courses. ✗
- **HippoRAG**: You have completed a total of **two** online courses on edX. ✗
- **RAPTOR**: To determine the total number of online courses you have completed, please provide the number of courses completed from each platform. No Answer

Case 2: Multi-session Question – Tomato and Cucumber Plants
Query: How many plants did I initially plant for tomatoes and cucumbers?
Ground-Truth Answer: 8
Responses of Different Methods:

- **MemGAS** : You initially planted 5 tomato plants and 3 cucumber plants. ✓
- **SeCom**: You initially planted 5 tomato plants. The conversation does not specify the number of cucumber plants you planted. ✗
- **MPC**: Based on previous conversations, you did not specify the number of tomato and cucumber plants. Please clarify how many of each you started with. No Answer
- **HippoRAG**: You didn't specify the exact number of plants you initially planted. Please provide that info for clarity. No Answer
- **RAPTOR**: Your message didn't specify the number. A good starting point is 1 tomato per 18–24 inches and cucumbers 12–18 inches apart. No Answer

Case 3: Temporal Reasoning – Meeting Emma
`Query:`How many days ago did I meet Emma?
`Ground-Truth Answer:` 9 days ago (10 days incl. the last day is acceptable)
`Responses of Different Methods:`

- **MemGAS** : You met Emma 9 days ago, on April 11, 2023. ✓
- **SeCom**: Please provide the date you met Emma. I can calculate how many days ago it was from April 20, 2023. No Answer
- **MPC**: The exact date of your meeting with Emma isn't available. If you provide it, I can compute the elapsed days. No Answer
- **HippoRAG**: There is no specific mention of meeting Emma or the date in the dialogs. Please provide the date. No Answer
- **RAPTOR**: No relevant information was found about Emma or the timeline. Please add more context. No Answer

Figure 12: Three representative queries with multi-turn or temporal reasoning, comparing the performance of various methods. MemGAS provides precise, context-integrated answers. The red text represents the key information that is relevant to the answer. ✓ means the response is correct while ✗ means it is wrong. No Answer refers to the response that does not provide the answer, which may be attributed to wrong retrieval.

Case of generate multi-granularity information

Origin session information

[user]

```
Write a sales email for a potential customer of a company that has a product
that enables energy saving ...  (Omitted for brevity)
```

[assistant]

```
Dear valued customer,
We at [Company Name] are excited to introduce our innovative ...  (Omitted
for brevity)
Sincerely,
[Your Name]
[Company Name]
```

Granularity Level of Summary

The user requested assistance in drafting a sales email aimed at potential customers for a company offering a product designed to enhance energy savings in industrial factories. Specifically, the product involves larger, more affordable pressure tanks for air compressors, allowing compressors to operate less frequently while maintaining necessary air pressure, thereby reducing energy consumption and operating costs.

Granularity level of keywords

- sales email
- potential customer
- energy saving
- industry factories
- pressure tanks
- air compressors
- cost-effective
- reduce energy consumption
- lower operating costs
- request a quote

Figure 13: An example of generating multi-granularity information using gpt-4o-mini. The blue text represents the key information summarized or extracted that is relevant to the conversation.

Case of generate multi-granularity information

Origin session information

[user]

```
Im´ thinking of taking an antique restoration class to learn how to fix up
some of the old pieces Ive´ inherited, like that beautiful vase I got from my
grandmother last month. Do you know of any good classes in the city?
```

[assistant]

```
I'll definitely look into those options. By the way, do you know how to
appraise the value of antique items? ... (Omitted for brevity)
```

[user]:

```
I'm thinking of taking an antique restoration class to learn how to fix up
some of the old pieces I've inherited, like that beautiful vase I got from my
grandmother last month.... (Omitted for brevity)
```

[assistant]

```
Appraising the value of antique items can be a fascinating and complex
process. While I'm not a professional appraiser ... (Omitted for brevity)
```

[user]:

...

[assistant]

...

Granularity Level of Summary

The user plans to retrieve their antique tea set, which they lent to their sister for her wedding. They express excitement about seeing the set again and appreciate the care and storage tips provided for antique china to maintain its condition. The user intends to ensure the tea set is safely back in their possession, and they also consider documenting its history and provenance for future preservation.

Granularity level of keywords

- antique tea set
- storage tips
- caring for china
- keep in good condition
- avoid stacking
- sturdy box
- line with tissue
- dry cool place
- ...

Figure 14: An example of generating multi-granularity information using gpt-4o-mini. The blue text represents the key information summarized or extracted that is relevant to the conversation.

Case of multi-granularity response filtering

Query

“I was looking back at our previous conversation about environmentally responsible supply chain practices, and I was wondering if you could remind me of the company you mentioned that’s doing a great job with sustainability?”

Answer

“Patagonia”

Before Filter

Memory 1:

Session-level information

... (Omitted for brevity)

[user]:

Can you give me an example of a company that has successfully implemented these environmentally responsible practices in its supply chain?

[assistant]

Yes, sure. Here’s an example:
Patagonia, an outdoor clothing and gear company, is known for its commitment to sustainability and environmental responsibility throughout its supply chain ... (Omitted for brevity)

Summary-level information

... (Omitted for brevity) **Patagonia** was highlighted as a prime example, showcasing its commitment to sustainability ...(Omitted for brevity)

Keyword-level information

- Patagonia
- environmentally responsible
- supply chain
- ...

Memory 2:

Session-level information

... (Omitted for brevity)

[user]:

I’m looking for some recommendations on eco-friendly laundry detergent. I’ve been trying to reduce my carbon ... (Omitted for brevity)

[assistant]

That a fantastic step towards reducing your environmental impact! Switching to an eco-friendly ... (Omitted for brevity)

Summary-level information

The user is considering trying a new grocery delivery service that partners with local farmers...(Omitted for brevity)

Keyword-level information

- meal planning
- grocery list
- start small
- ...

After Filter

The two companies mentioned that prioritize employee safety and well-being like Triumvirate are:
1. Patagonia - Known for its commitment to environmental conservation and prioritizing employee well-being, offering flexible work arrangements and generous leave policies.
2. Southwest Airlines - Emphasizes employee care, providing top-notch training and benefits, resulting in a positive work environment and strong focus on safety and customer service.

Figure 15: An example of multi-granularity information filtering for retrieved conversations. The red text represents the key information that is relevant to the answer.

Case of multi-granularity response filtering

Query
"I was going through our previous conversation and I was wondering if you could remind me of the two companies you mentioned that prioritize employee safety and well-being like Triumvirate?"
Answer
"Patagonia and Southwest Airlines."

Before Filter

Memory 1:
Session-level information
... (Omitted for brevity)
[user]:
For sure! It's always a win-win when companies take care of their employees. It makes me feel better about supporting them as a customer. Do you know of any other companies that prioritize ... (Omitted for brevity)

[assistant]

I can give you an example of two companies that prioritize the safety and well-being of their employees like Triumvirate does:
1. Patagonia ... (Omitted for brevity)
2. Southwest Airlines ... (Omitted for brevity)
Summary-level information
... (Omitted for brevity) to which the AI cited Patagonia and Southwest Airlines as examples of organizations that prioritize employee safety and well-being ...(Omitted for brevity)
Keyword-level information
- Triumvirate
- employee safety
- tangible improvements
- ...

Memory 2:
Session-level information
... (Omitted for brevity)
[user]:
I'm thinking of trying out that new grocery delivery service that partners with local farmers. ... (Omitted for brevity)

[assistant]

That sounds like a great idea! I'm happy to help you with that ... (Omitted for brevity)
Summary-level information
The user is focused on making eco-friendly changes in their kitchen, specifically looking for sustainable alternatives to kitchen utensils, ...(Omitted for brevity)
Keyword-level information
- eco-friendly kitchen makeover
- sustainable kitchen utensils
- bamboo utensils
- ...

After Filter

The company mentioned that is doing a great job with sustainability is Patagonia. They are known for their commitment to environmentally responsible practices throughout their supply chain, including sustainable sourcing, green transportation, packaging optimization, waste reduction, and compliance with environmental regulations.

Figure 16: An example of multi-granularity information filtering for retrieved conversations. The red text represents the key information that is relevant to the answer.